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HPCL ASSIGNMENT 3

QUESTIONS-

Q.1) Write a C program that adds two integer arrays, A and B, each containing N elements, and stores the result in array C.

Q.2) Write a C program that calculates the sum and average of N floating-point numbers stored in an array. Dynamically allocate the array using malloc.

Initialize the array with random numbers.

Q.3) Write a C program that finds the largest value in an integer array and reports its index.

Q.4) Write a C program that multiplies two square matrices, A and B, of size N × N and stores the result in matrix C.

Q.5) Write a C program that counts how many prime numbers exist between 2 and a user-provided upper limit N.

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Q.1-

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```
  GNU nano 7.2
#include <stdio.h>

#define N 5

int main()
{
    int A[N] = {1, 2, 3, 4, 5};
    int B[N] = {10, 20, 30, 40, 50};
    int C[N];

    for (int i = 0; i < N; i++) {
        C[i] = A[i] + B[i];
    }

    printf("Result array C:\n");

    for (int i = 0; i < N; i++) {
        printf("%d ", C[i]);
    }

    printf("\n");

    return 0;
}
```

O/P—

Cloud Shell Editor

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```
armaanpatnaik2004@cloudshell:~$ nano array_add.c
armaanpatnaik2004@cloudshell:~$ gcc array_add.c -o array_add
./array_add
Result array C:
11 22 33 44 55
armaanpatnaik2004@cloudshell:~$
```

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Q.2-

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```
  GNU nano 7.2
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

int main()
{
    int N;
    float sum = 0;

    printf("Enter number of elements: ");
    scanf("%d", &N);

    float *arr = malloc(N * sizeof(float));

    srand(time(NULL));

    printf("Numbers:\n");

    for (int i = 0; i < N; i++) {
        arr[i] = (rand() % 100) / 10.0;
        sum += arr[i];

        printf("%.1f ", arr[i]);
    }

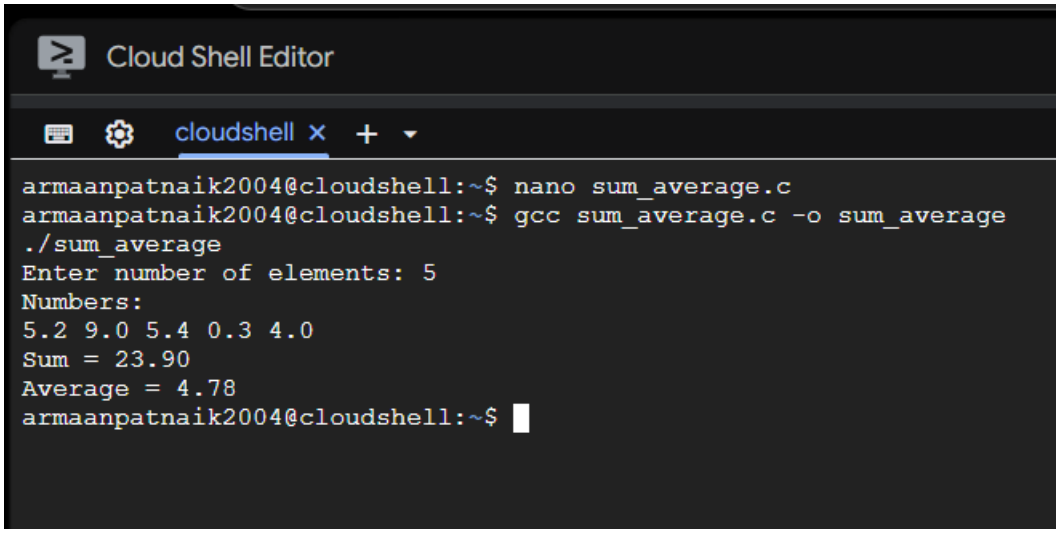
    printf("\nSum = %.2f\n", sum);
    printf("Average = %.2f\n", sum / N);

    free(arr);

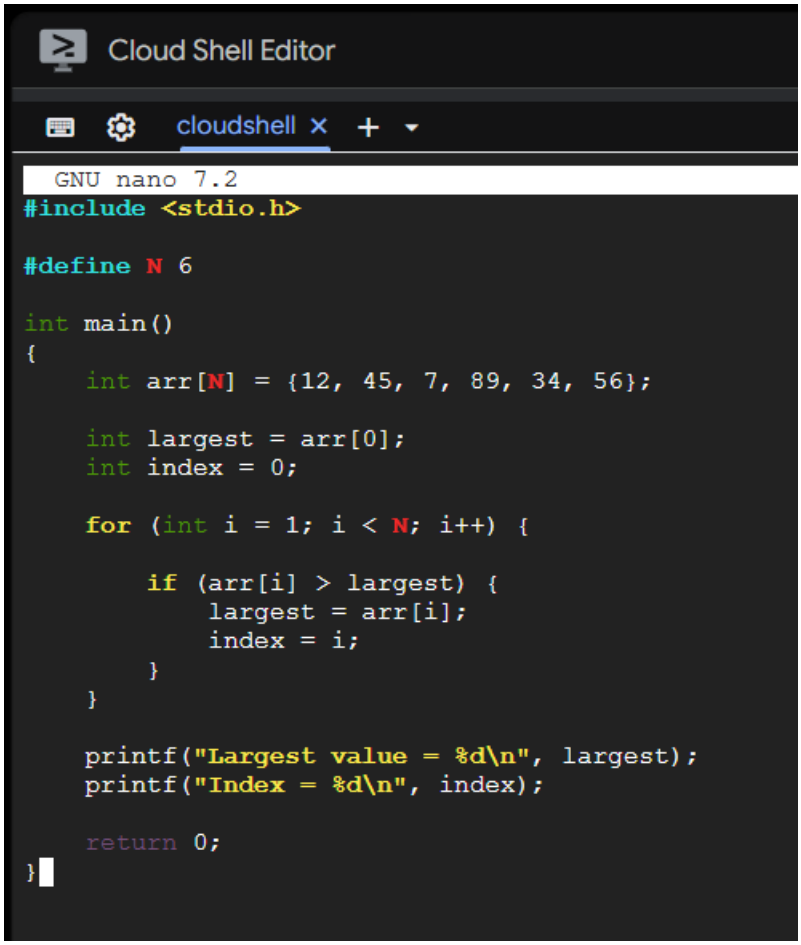
    return 0;
}
```

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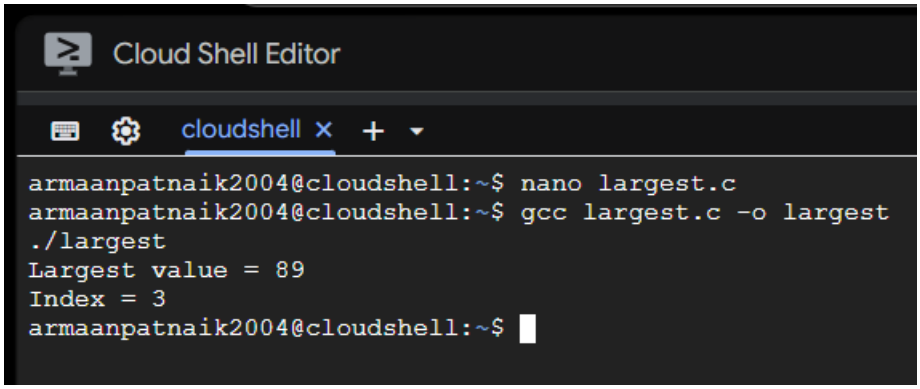


q.3 –



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Q.4—

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```
 GNU nano 7.2
#include <stdio.h>

#define N 3

int main()
{
    int A[N][N] = {
        {1, 2, 3},
        {4, 5, 6},
        {7, 8, 9}
    };

    int B[N][N] = {
        {9, 8, 7},
        {6, 5, 4},
        {3, 2, 1}
    };

    int C[N][N] = {0};

    for (int i = 0; i < N; i++) {

        for (int j = 0; j < N; j++) {

            for (int k = 0; k < N; k++) {
                C[i][j] += A[i][k] * B[k][j];
            }
        }
    }

    printf("Result Matrix:\n");

    for (int i = 0; i < N; i++) {

        for (int j = 0; j < N; j++) {
            printf("%d ", C[i][j]);
        }

        printf("\n");
    }

    return 0;
}
```

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```
armaanpatnaik2004@cloudshell:~$ nano matrix_multiply.c
armaanpatnaik2004@cloudshell:~$ gcc matrix_multiply.c -o matrix_multiply
./matrix_multiply
Result Matrix:
30 24 18
84 69 54
138 114 90
armaanpatnaik2004@cloudshell:~$
```

q.5—

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Cloud Shell Editor

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```
  GNU nano 7.2
#include <stdio.h>

int main()
{
    int N, count = 0;

    printf("Enter upper limit: ");
    scanf("%d", &N);

    for (int i = 2; i <= N; i++) {

        int prime = 1;

        for (int j = 2; j * j <= i; j++) {

            if (i % j == 0) {
                prime = 0;
                break;
            }
        }

        if (prime) {
            count++;
        }
    }

    printf("Number of prime numbers between 2 and %d = %d\n",
           N, count);

    return 0;
}
```

o/p-

Cloud Shell Editor

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```
armaanpatnaik2004@cloudshell:~$ nano prime_count.c
armaanpatnaik2004@cloudshell:~$ gcc prime_count.c -o prime_count
./prime_count
Enter upper limit: 5
Number of prime numbers between 2 and 5 = 3
armaanpatnaik2004@cloudshell:~$
```